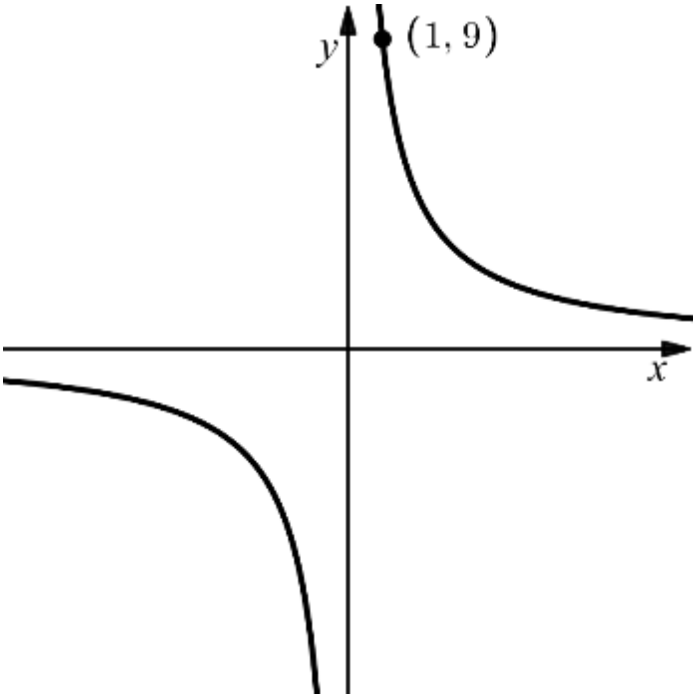
1.24 Graph Sketching: Rectangular Hyperbolas

Worked Solutions

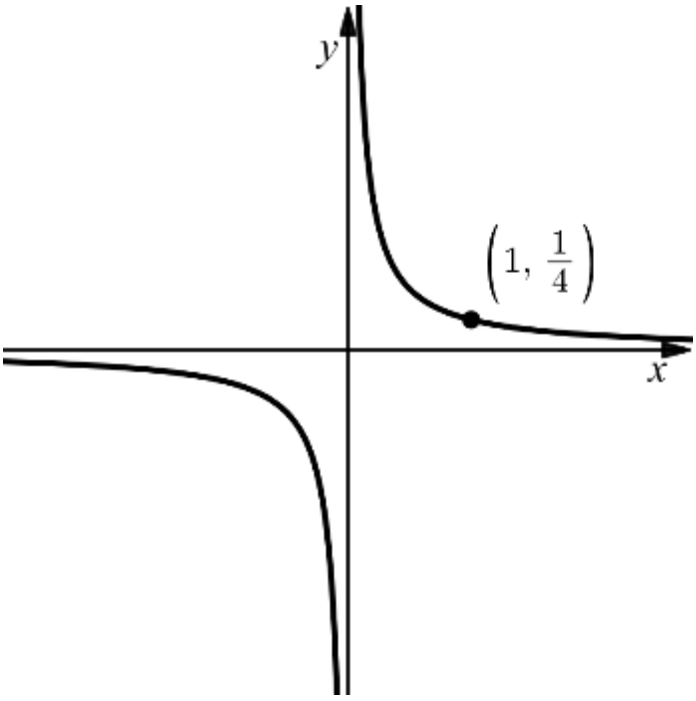
1 a)



Asymptotes: $x = 0$, $y = 0$

Lines of symmetry: $y = -x$ and $y = x$

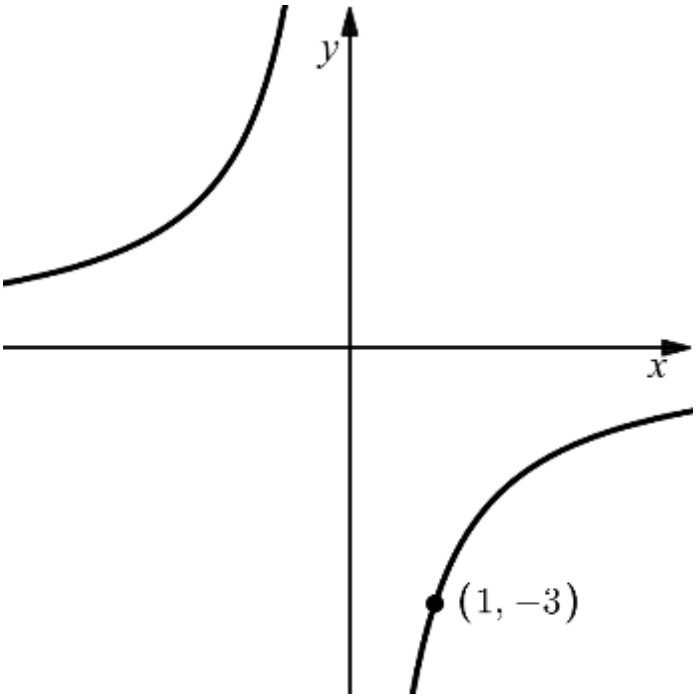
1 b)



Asymptotes: $x = 0$, $y = 0$

Lines of symmetry: $y = -x$ and $y = x$

1 c)



Asymptotes: $x = 0$, $y = 0$

Lines of symmetry: $y = -x$ and $y = x$

2 a)

$$\begin{aligned} xy &= k \\ 2 \times 8 &= 16 \\ \therefore xy &= 16 \\ a &= 16 \end{aligned}$$

2 b)

$$\begin{aligned} xy &= k \\ 2 \times -4.9 &= -9 \\ \therefore xy &= -9 \\ a &= -9 \end{aligned}$$

3)

$x^2 - y^2 = k$ has asymptotes $y = x$ and $y = -x$

Considering $m_1 m_2$ then

$$1 \times -1 = -1$$

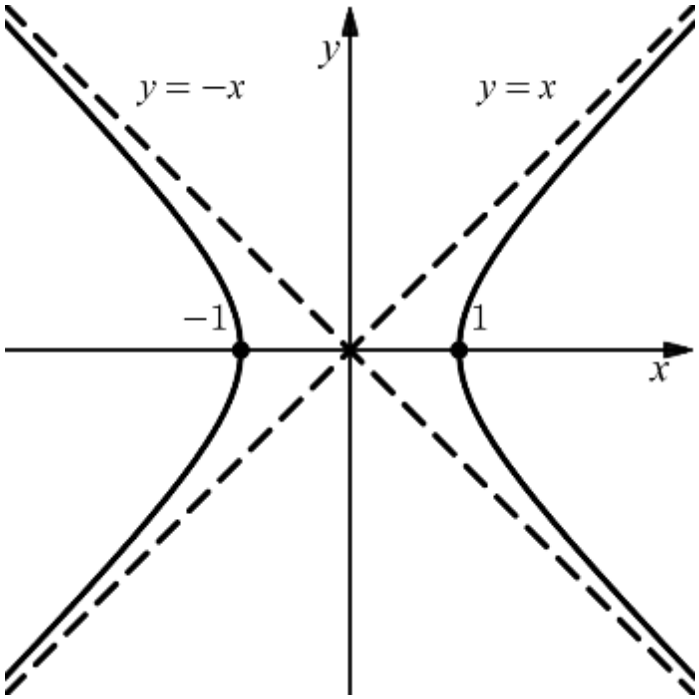
$\therefore$ the asymptotes are perpendicular

4 a)

$x^2 - y^2 = 1 \Rightarrow \frac{x^2}{1} - \frac{y^2}{1} = 1$

When $y = 0$, $x = \pm 1$

Asymptotes: $y = x$, $y = -x$

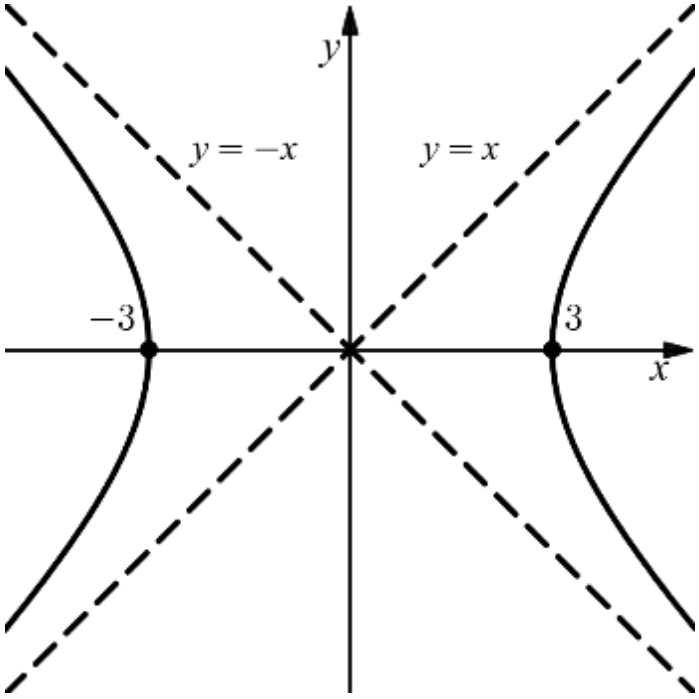


4 b)

$x^2 - y^2 = 9 \Rightarrow \frac{x^2}{9} - \frac{y^2}{9} = 1$

When $y = 0$, $x = \pm 3$

Asymptotes: $y = x$, $y = -x$

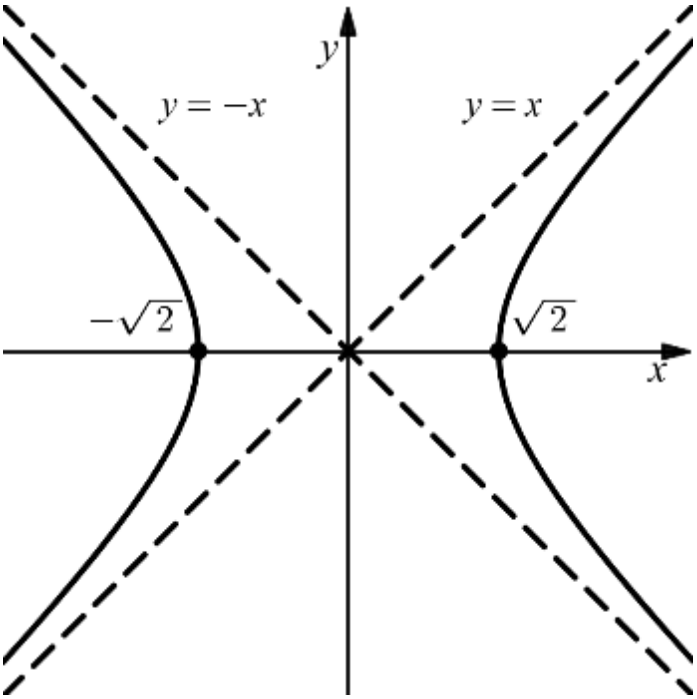


4 c)

$\frac{x^2}{2} - \frac{y^2}{2} = 1$

When $y = 0$, $x = \pm\sqrt{2}$

Asymptotes: $y = x$, $y = -x$

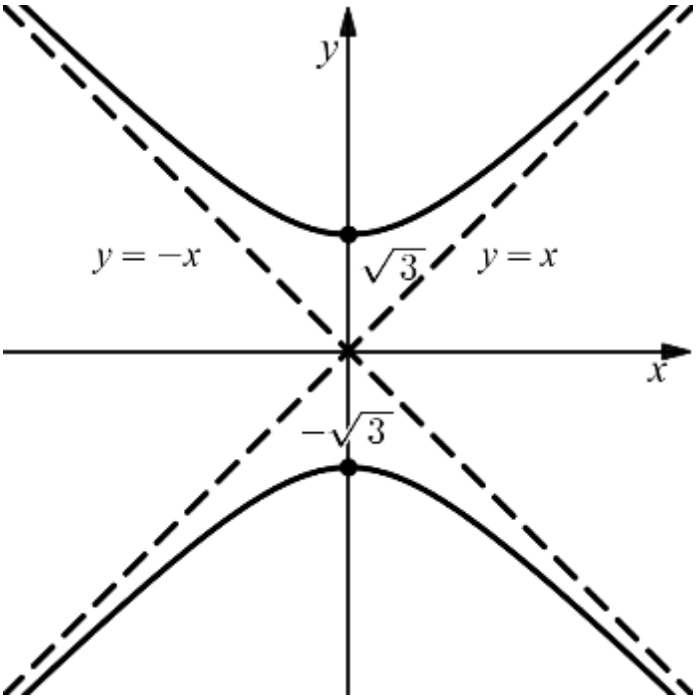


4 d)

$y^2 - x^2 = 3$

When $x = 0$, $y = \pm\sqrt{3}$

Asymptotes: $y = x$, $y = -x$



5 a)

$\frac{x^2}{4} - \frac{y^2}{4} = 1 \quad \therefore k = 4$

5 b)

$\frac{x^2}{8} - \frac{y^2}{8} = 1 \quad \therefore k = 8$

5 c)

$\frac{x^2}{5} - \frac{y^2}{5} = -1 \Rightarrow x^2 - y^2 = -5$

$\therefore k = -5$

6 a)

$k = a^2 - b$

6 b)

$x = \sqrt{a^2 - b}$ or $x = -\sqrt{a^2 - b}$